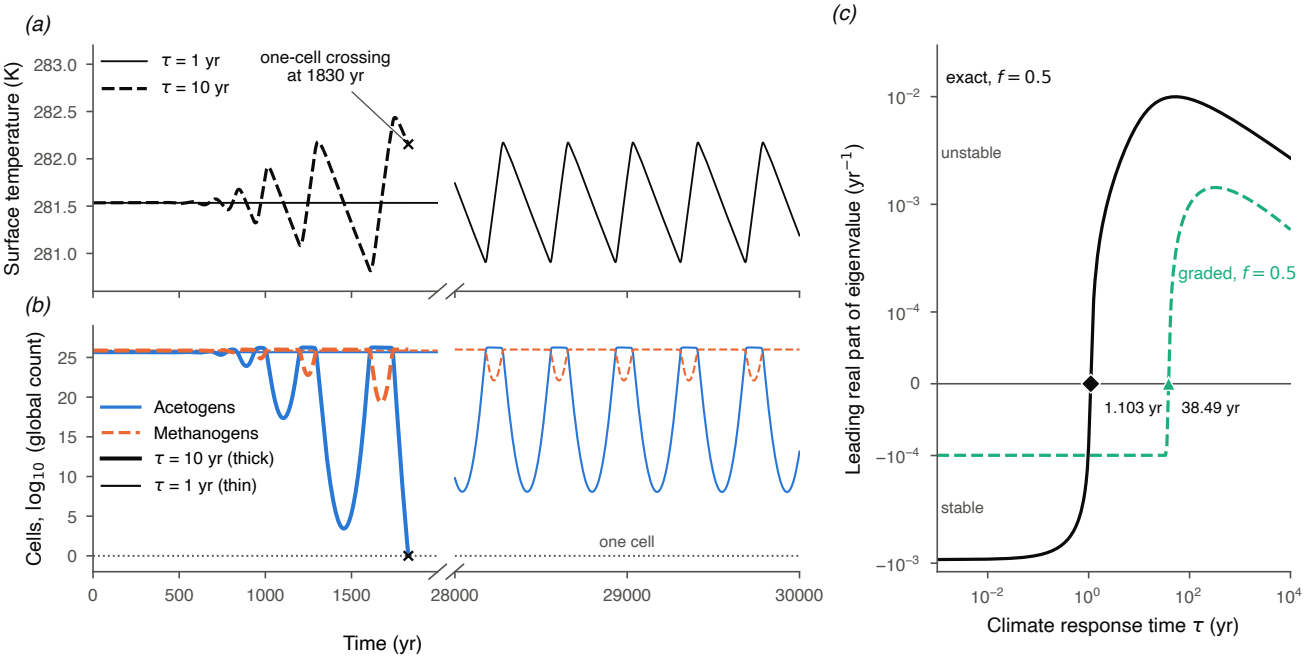
(a)
τ = 1 yr
τ = 10 yr
one-cell crossing
at 1830 yr
Surface temperature (K)
283.0
282.5
282.0
281.5
281.0
(b)
Acetogens
Methanogens
τ = 10 yr (thick)
τ = 1 yr (thin)
Cells, log10 (global count)
25
20
15
10
5
0
one cell
0
500
1000
1500
28000
29000
30000
Time (yr)
(c)
exact, f = 0.5
unstable
graded, f = 0.5
1.103 yr
38.49 yr
stable
Leading real part of eigenvalue (yr−1)
10−2
10−3
10−4
0
−10−4
−10−3
10−2
100
102
104
Climate response time τ (yr)